MogScan: Real-Time, Multi-Person, Explainable Facial-Aesthetics Scoring on a Commodity Webcam

Final Project — Startup / Product Track (Modified YC Application)

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Abstract

Face-rating ("looksmaxxing") apps are some of the most downloaded AI consumer products of the last two years. Omoggle, a webcam "mog battle" site, recently spread across Twitch. Current products, however, turn ratings into competitions or shallow category scores, give a number without a credible explanation, usually upload the user's selfie to a server, and do not show that their scores agree with real human judgements. We built **MogScan**, a browser app that takes a live USB webcam feed or an uploaded photo and gives each face (up to six), a live PSL score (1–8), a community tier (Sub5 … HTN, Chadlite, Chad), and a ranked list of the measured facial features that raise or lower that score, from canthal tilt and FWHR to orbital vector, skin texture and jawline definition. Processing runs fully on-device. The pipeline frontalizes MediaPipe's 478-landmark face mesh and computes 143 inputs: 58 geometric, 3D-depth and symmetry measurements, 14 whole-face shape features, 20 skin, colouring, hair and leanness features from the pixels, and 51 expression blendshapes. It then combines an explainable, sign-constrained ridge model over all of them with a fine-tuned MobileNetV3. On the official SCUT-FBP5500 60/40 split, our fused model reaches Pearson $r = 0.920$ against the mean of 60 human raters. That is higher than the dataset's published ResNeXt-50 baseline ($r = 0.8777$), and the model runs in real time in a browser tab. We also report subgroup bias, which "looksmax rules" do and do not match human ratings, and a user study with **[run study]** participants.

[Insert figures/hero.png: MogScan screenshot of one consenting person with the card showing]

Figure 1: MogScan scoring a live USB-webcam feed. After a two-second scan the face gets a locked PSL score and tier. The side card shows four pillar sub-scores and the top three strengths and things to improve; every measurement, next to the average of the top-20%-rated faces, and every model weight are one click away.

1 Introduction

Motivation. "Looksmaxxing" is the online subculture of measuring and improving one's appearance. It has turned into a large consumer AI market. Industry reporting put the leading face-rating app, Umax, at over 7 million downloads and roughly $500K per month in subscription revenue by mid-2024 [19]. In 2026 *Omoggle*, an Omegle-style site that pairs two strangers on webcam and declares a "mogger", spread across Twitch after streamers such as xQc and Asmongold played it on stream [6]. The demand is clear. People want to know where they stand on the community's PSL scale and, more importantly, *why*. Omoggle turns this into a competition between strangers. We think most people want the opposite: a private, detailed assessment of their own face.

The products serving this demand have four recurring problems: (1) **shallow measurement**: selfie apps rate one photo on a handful of categories, and Omoggle reduces a face to a duel result; (2) **unexplained numbers**: the scores are opaque, or the "feature analysis" is generic text; (3) **privacy**: most apps upload a face photo to a server; (4) **no validation**: we found no competitor that reports how well its scores agree with human raters. Coverage of Omoggle notes that camera angle and distance easily flip its results [6].

What we built. MogScan is a web app with two inputs: a live camera (any USB webcam, picked from the camera menu) or an uploaded photo. Every face in view is scored live (Figure 1). For each person MogScan shows:

- a **PSL score** (1–8), the percentile, and the community tier (Sub3, Sub5, LTN, MTN, HTN, Chadlite, Chad, Adamlite+; female variants LTB/MTB/HTB, Stacylite, Stacy);
- the **top three strengths and weaknesses**, each with its measured value, its effect on the score in PSL points, the average of top-rated faces, and the community's reference range;
- a **glow-up plan**: non-surgical actions (skincare, contouring the jawline, filling in brows, grooming, losing facial puffiness, …) ranked by the score gain the model predicts for this face, each with a predicted score, plus an overall "potential" score (§4.8);
- five **pillar sub-scores** modelled on the looksmax community's breakdown (Harmony, Dimorphism, Angularity/leanness, Softmax, Expression);
- all **143 inputs** on demand, grouped, each with its value, its learned weights and its effect on the score, available on demand.

The interface is deliberately minimal. There is no competitive mode, ranking or leaderboard: if several people are in view, each is scored on their own. Study builds add a Record button that exports an anonymized numeric session log for our user study. Nothing leaves the device.

Contributions.

(1) A real-time, multi-person, fully client-side pipeline: landmarks, tracking, pose gating, frontalization, metrics, two models, and fusion, running at camera frame rate in a browser tab.
(2) An *explainable* feature model over 143 inputs whose per-feature attributions have signs that agree with the evidence (or, for a few features, with an uncontroversial prior), obtained with a sign-constrained ridge regression. We measure the accuracy cost of this constraint and what each of the 14 feature groups adds.
(3) An evaluation on SCUT-FBP5500: accuracy against human raters, subgroup bias, and a data-driven check of popular looksmax rules.
(4) Evidence from real users: a usability and value study (§5.5).

2 Modified YC Application

Describe what your company does in 50 characters or less. Private, explainable face ratings, live.

What is your company going to make? A camera-or-photo product that rates a face in real time and explains the score with 143 measured features. The first product is the free web app described in this paper. Next comes a mobile app with progress tracking and a side-profile scan.

Why did you pick this idea? Why now? Two things came together in 2026. First, demand: Omoggle and the selfie raters showed that live face rating has mass appeal [6, 19], but both frame it as competition or a bare number. Second, capability: browser ML (MediaPipe Tasks [14], ONNX Runtime Web) is now fast enough to run a 478-point face mesh for six people *and* a CNN on a laptop, without a server. That makes "on-device, nothing uploaded" free for us to offer and costly for server-based competitors to match.

Who are your users and how do you know they need it? (a) 18–25-year-olds in the looksmaxxing community, who already pay $4–$10 per week for selfie raters [12, 20]; (b) people tracking the effect of grooming, skin care or weight loss on their face. Evidence from our own study is in §5.5: SUS **[run study]**, NPS **[run study]**, **[run study]**% found the sub-scores useful, and **[run study]**% said they would pay for an extended version.

Who are your competitors? What do you understand that they don't? See Table 1. Our insight is that *explanations are the product.* Users do not just want a number. They want to know *which* features help or hurt them. Explanations only earn trust if they agree with the evidence, so we validate the score against human raters and build the explanation model so that its signs cannot contradict the data (§4.5).

How will you make money? Free live web app, including the glow-up plan, as the acquisition channel. **MogScan Pro** at $4.99/month adds history, progress tracking against the plan, and a side-profile scan. That is cheaper than the $4–$10 *per week* the incumbents charge [12]. Our marginal compute cost per user is close to zero because inference runs on the user's device.

Traction so far. Working product (this paper), an accuracy result above the dataset's published baseline, and a user study with **[run study]** participants (§5.5).

What is the biggest risk? Ethical and reputational harm: body-image damage, use on minors, and rating people without consent (§6). A practical risk is data licensing: SCUT-FBP5500 is licensed for non-commercial research, so a commercial release needs a model retrained on licensed or self-collected ratings.

3 Related Work

3.1 Competing products

Omoggle [18] is the closest live product. It runs a 468-landmark face mesh in the browser and turns 10-second 1v1 duels into an ELO ladder (Molecule … Chadlite, Chad, Slayer). We deliberately avoid that format: MogScan has no duels, rankings or leaderboards. Omoggle's scoring is a black box, and press coverage notes that camera angle, distance and props easily change results [6]. MogScan addresses this with *pose gating* and *frontalization* (§4.3). The selfie apps (Umax, LooksMax AI, and dozens of PSL raters) show strong demand, but they put results behind paywalls and upload photos [12, 19, 20]. A parental-monitoring review rates PSL apps 18+ and warns about obsessive checking [1]; we take that seriously in §6. Qoves Studio sells human-written facial-aesthetics reports at a much higher price point [22].

3.2 Research

Facial beauty prediction (FBP). Early work regressed attractiveness on hand-crafted geometric ratios and found that a few proportions explain a large share of the variance in ratings [7, 9, 13]. SCUT-FBP [23] and SCUT-FBP5500 [16] made deep FBP standard. On the 60/40 split, the SCUT-FBP5500 paper reports Pearson correlations of 0.8298 (AlexNet), 0.8513 (ResNet-18) and 0.8777 (ResNeXt-50), and 0.8997 for ResNeXt-50 under 5-fold cross-validation. Recent large hybrid models report around 0.92 [2]. These works target offline accuracy on single photos. We target real-time, multi-person, *explained* predictions on a webcam, with a model small enough for a browser.

Psychology of attractiveness. Meta-analyses find strong agreement between raters on attractiveness within and across cultures [15]. That agreement is what makes a learned predictor possible. Averageness, symmetry and sexual dimorphism are the best-supported cues [21]. Facial width-to-height ratio (FWHR) is linked to perceived dominance [4]. Looksmax heuristics such as "positive canthal tilt" or a "1:2 philtrum-to-chin ratio" mostly come from forums, not studies. We test them against human ratings directly (Figure 5).

Face tracking. We use MediaPipe's FaceLandmarker (478 landmarks including irises, plus a facial transformation matrix) [8, 14] and MobileNetV3 [11] for the CNN.

Explainability and fairness. Additive attributions such as SHAP [17] decompose a prediction into per-feature contributions. Our geometric model is additive by construction, so its attributions are exact. Commercial face analysis has documented subgroup disparities [3], and debiasing for aesthetic prediction has been proposed [5]. We report per-subgroup error in Table 6.

Table 1: Competitors and alternatives. Prices come from public listings and third-party reports (Oct. 2026) and vary by region and listing.

	Omoggle [6, 18]	**UMAX / LooksMax AI** [12, 19]	**PSL rating apps** [1, 20]	**Qoves Studio** [22]	**MogScan (ours)**
Input	Live webcam, 1v1 with a stranger	One selfie	One selfie	Uploaded photos	Live webcam (up to 6 faces) **or** photo upload
Output	0–10 score, win/loss, ELO rank	Overall + sub-scores (jaw, skin, …)	PSL score + tier	Written report	PSL, tier, percentile, 5 pillar sub-scores, 143 measured inputs with weights and PSL effect
Explanation	Mentions symmetry, canthal tilt, jaw; none shown per user	Category scores	Category scores	Human-written analysis	Per-feature attribution from a validated, sign-constrained model
Validated vs. human raters	Not reported; reportedly sensitive to angle and props	Not reported	Not reported	Not reported	Pearson $r = 0.920$ on a held-out test split
Privacy	Claims on-device	Server upload	Server upload	Server upload	On-device, nothing stored
Price	Free	≈$4–10 / week	Subscription	≈$20–200 one-time	Free; Pro $4.99 / month (planned)

4 Implementation

4.1 Design process and ideation

Our initial plan was a direct Omoggle-style design: a face mesh, looksmax ratios, and a score from hand-picked "ideal" ranges for each ratio. Four findings during development reshaped it:

(1) **Hand-picked ideals are arbitrary.** Their weights had no justification, and some community ideals do not match the measurements MediaPipe's landmarks produce. We switched to *learning* the weights from 5,500 faces rated by 60 people [16].
(2) **Head pose leaks into every metric.** On the dataset, our symmetry metric correlated $r = 0.97$ with |yaw|, so it was measuring head turn, not the face. Frontalization reduced this to $r = 0.67$, and pose gating in the app handles the rest (§4.3).
(3) **Unconstrained explanations contradict the data.** The metrics are correlated with each other (e.g. jaw : cheekbone width vs. jaw angle, $r = -0.75$; FWHR vs. face length : width, $r = -0.68$). An ordinary ridge regression therefore gave FWHR a large *positive* weight (+0.26 for male and +0.46 for female faces, in rating units per SD), even though FWHR on its own correlates *negatively* with ratings ($r \approx -0.25$). Users with a high FWHR would have been told it was a strength while the raw data says the opposite. We now constrain the signs (§4.5).
(4) **Camera pose ≠ apparent pose.** Our first pose gate used MediaPipe's transformation matrix. It rejected a frontal face near the edge of a multi-person frame, because the matrix includes the off-axis viewing angle. We now gate on the apparent pose from frontalization, which is what the metrics actually depend on.

For the interface we chose a browser app over a native OpenCV window. A browser tab runs on any laptop with any USB camera, can be shared as a link, and keeps everything on-device. Live cards update five times per second, while the overlay is drawn on every frame.

4.2 Landmarks, tracking and pose gating

MediaPipe FaceLandmarker runs in `VIDEO` mode on the GPU delegate, falling back to CPU, with up to six faces, and returns 478 normalized 3D landmarks per face. We read the *apparent* head pose, yaw ψ and pitch, from the rotation that aligns the face to our frontal template (§4.3). We do not use MediaPipe's facial transformation matrix for this. That matrix measures rotation relative to the camera ray, so in testing a perfectly frontal face near the edge of a wide frame read as $-17°$ of yaw. The Procrustes yaw correlates $r = 0.99$ with MediaPipe's yaw on the centered dataset images, at about 1.2× the scale. A greedy IoU tracker (match threshold 0.25, 1 s time-to-live) keeps a stable ID per person. Each person is *scanned* once: features are collected *only* while the face is near-frontal ($|\psi| < 14°$, $|\text{pitch}| < 20°$) until there are 60 frames (about 2 s) and 12 CNN and pixel samples. The result then *locks* to the per-feature median of the scan, so the score, the pillars and the strengths and weaknesses no longer jitter from frame to frame. A Rescan button starts over. An earlier version smoothed every frame with a moving average; users found the constantly shifting score and re-ordering strengths hard to trust. This is a direct fix for the angle sensitivity reported for Omoggle.

4.3 Frontalization

Landmarks are converted to pixel-space 3D (xW, yH, zW), centered, and scaled to unit RMS radius. We then find the rotation that best aligns them to a mean frontal template using Horn's closed-form quaternion method [10]: the top eigenvector of a 4×4 symmetric matrix built from the cross-covariance. The template is a generalized-Procrustes mean of 1,257 near-frontal *training* faces. Metrics are computed on the x, y coordinates of the aligned shape, after removing residual roll so that the line between the pupils is horizontal; depth features use the aligned z coordinate. The browser uses a Jacobi eigen-solver port of the same algorithm. Unit tests check every geometric, shape and pixel feature against the Python reference

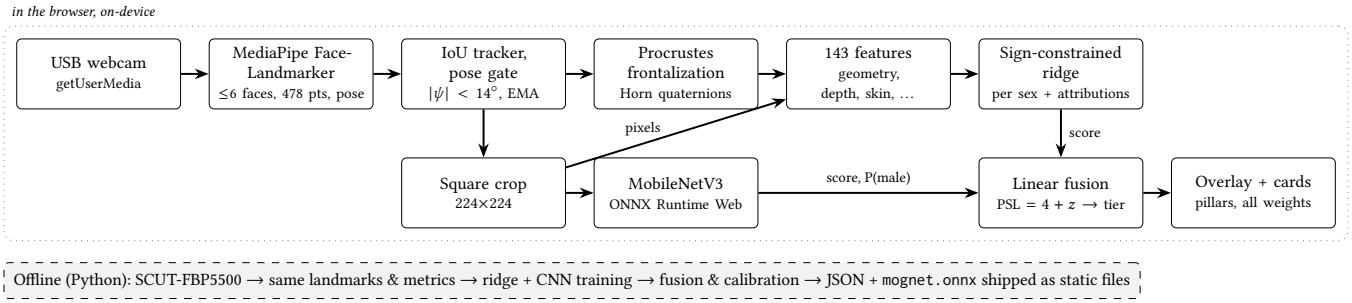


Figure 2: System architecture. The browser pipeline (top) and the offline training pipeline (bottom) share the same landmark model, crop rule and metric code. The TypeScript metric implementation is checked against the Python reference in unit tests.

within 10^{-3} on five faces, including the most strongly turned faces in the dataset.

4.4 Features

We set out to measure every factor in popular looksmax rating breakdowns that a single frontal photo can support. All lengths are normalized by the interpupillary distance (IPD), the bizygomatic width or the face length, so no feature depends on how far someone is from the camera. The 143 inputs fall into 14 groups:

- **Eyes, brows, nose, mouth, jaw, proportions** (58 with depth and symmetry): the 15 classic ratios of Table 2 plus canthal asymmetry, intercanthal : eye width, iris : eye width, upper and lower scleral show, brow tilt, thickness, length and gap, nose length, bridge and tip width (bulbousness), mouth : IPD, lip fullness, cupid's-bow depth, mouth-corner tilt, chin height and width, chin taper, ramus length, jaw width : face length, upper third, temple width, cheekbone height and lower-face fullness.
- **Profile depth**: a pseudo-profile from the frontalized mesh's z coordinate gives nasal, chin, lip (Ricketts E-line), cheekbone and midface projection, brow-ridge depth, orbital vector, forehead slope, facial convexity and overall face depth.
- **Symmetry**: overall and per region (eyes, brows, nose, mouth, jaw).
- **Whole-face shape**: averageness (RMS distance from the mean face), a sexual-dimorphism score (projection on the male–female mean shape difference) and 12 shape principal components.
- **Pixels** (20, from a 160×160 crop): skin brightness, tone unevenness, texture, redness, undertone and under-eye darkness; lip, eye, brow and lash-line contrast, sclera brightness and eye colour; hair darkness and texture and facial hair; and leanness cues (jawline shadow, cheek hollowness, neck width).
- **Expression**: MediaPipe's 51 blendshapes (smile, squint, brow raise, …).

Physique, shoulder width, body fat below the face and clothing, which appear in some looksmax checklists, cannot be measured from a face crop and have no ratings in the training data, so we leave them out. Table 2 lists the landmark definitions of the 15 classic ratios; the code documents the rest.

Table 2: The 15 classic looksmax ratios (MediaPipe landmark indices in parentheses).

Metric	Definition
Canthal tilt	mean angle of inner→outer canthus vs. pupil line (133–33, 362–263); + = outer higher
Eye spacing (ESR)	IPD (468–473) / bizygomatic width (234–454)
FWHR	bizygomatic width / height glabella–upper lip (9–0)
Midface ratio	pupil line–upper lip height / IPD
Lower third	(subnasale–menton) / (glabella–menton) (2, 152, 9)
Chin : philtrum	(lower lip–menton) / (subnasale–upper lip) (17–152, 2–0)
Nose : intercanthal	alar width (129–358) / intercanthal width (133–362)
Mouth : nose	mouth width (61–291) / alar width
Jaw : cheekbone	bigonial (172–397) / bizygomatic width
Lower : upper lip	lower-lip height (14–17) / upper-lip height (0–13)
Eye openness	mean lid aperture / eye width (159–145, 386–374)
Brow–eye distance	brow (105, 334) to upper lid, / IPD
Face length : width	forehead–menton (10–152) / bizygomatic width
Asymmetry	mean mirror error of 13 bilateral pairs about the midline, / IPD
Jaw angle	projected angle zygion–gonion–menton (234–172–152, 454–397–152)

4.5 Explainable feature model

For each sex we standardize every feature (z_i, clipped to ±4) and fit

$$\hat{y}_{\text{geo}} = b + \sum_{i=1}^{143} \left(\beta_i z_i + \gamma_i (z_i^2 - 1)\right) + \text{pose terms},$$

where the pose terms ($|\psi|$, |pitch − median|) absorb any remaining pose effects during training and are set to zero at inference. In the *sign-constrained* variant, $\text{sign}(\beta_i)$ must match the sign of metric i's marginal correlation with the rating, and $\gamma_i \le 0$, so any curvature can only express an "ideal range" (an inverted U). For 16 features whose good direction is not in dispute (asymmetry, uneven or rough skin, a bulbous nose tip, dark circles, lower-face puffiness, jawline definition, …) the sign comes from that prior instead. Without it, confounds in the photos would make the app call "patchy skin" or "asymmetry" a strength; if the data disagree with the prior, the weight goes to zero. We solve it as non-negative ridge regression on a re-signed design matrix, with α chosen by 5-fold cross-validation. Because the model is additive, the contribution $c_i = \beta_i z_i + \gamma_i (z_i^2 - 1)$ is exact and has zero mean over the training set, so it reads as "points above or below an average face." The UI shows c_i/σ_y in PSL units, the top three positive and negative contributions, every feature's β_i, γ_i and importance (the SD of c_i), and pillar sub-scores

Table 3: Effect of simulated actions on the full MogScan score for 300 held-out test faces: share of faces whose score rose, and mean change in PSL.

	Male faces		Female faces	
Action	Raised	ΔPSL	Raised	ΔPSL
Fill in brows	95%	+0.163	93%	+0.192
Skincare (smoother, even skin)	85%	+0.115	74%	+0.095
Define lash line	82%	+0.055	71%	+0.050
Jawline contour	66%	+0.029	50%	+0.002
Self-tanner / bronzer	65%	+0.027	54%	+0.014
Brighter sclera	62%	+0.006	59%	+0.010
Reduce redness	35%	-0.005	42%	-0.005
Brighten under-eyes	34%	-0.023	31%	-0.029
Stubble	44%	-0.034	–	–
Lip colour	30%	-0.041	45%	-0.015
Cheekbone contour	19%	-0.045	31%	-0.038
All of the above	82%	+0.294	81%	+0.222

that sum the contributions of each pillar's feature groups. The "top-tier average" shown for each metric is that metric's mean over the top 20% of rated training faces of the same sex.

4.6 CNN

We fine-tune an ImageNet-pretrained MobileNetV3-Large [11] with a two-output head (beauty score, sex logit; loss = smooth-L1 + 0.2×BCE). Crops are square boxes 1.35× the landmark bounding box, black-padded, resized to 224×224. The browser uses exactly the same rule, so training and live inputs are framed the same way. Training used 85% of the official training split, with 15% held out for validation. Settings: AdamW, one-cycle learning rate with a peak of 3×10^{-4}, batch 32, 25 epochs, flips, ±8° affine and color jitter. We kept the epoch with the best validation correlation ($r = 0.915$). The model is exported to ONNX (opset 17, 13.9 MB) and runs in ONNX Runtime Web (WASM). A round-robin scheduler sends one still-scanning face to the CNN every 120 ms.

4.7 Fusion and PSL calibration

A linear regression on the validation set combines the two models: $\hat{y} = 0.95\,\hat{y}_{\text{cnn}} + 0.13\,\hat{y}_{\text{geo}} + b$. We map to PSL with $\text{PSL} = 4 + (\hat{y} - \mu)/\sigma$, where $\mu = 2.99$ and $\sigma = 0.69$ are the mean and SD of the human ratings, clipped to $[1, 8]$. With this mapping the community tier cut-offs land on familiar percentiles: MTN covers the middle (3.75–4.5), HTN is roughly the top 31%, Chadlite the top 10% (≥ 5.25), and Chad the top 2.3% (≥ 6). The detected sex chooses the geometric model and the tier names, and users can correct it by clicking the sex badge on their card.

4.8 Glow-up plan

The plan suggests non-surgical actions and predicts the score after each one. A prediction is only useful if doing the action really moves the score the app shows, so we *measured* it. For 300 held-out test faces (150 per sex) we simulated each cosmetic action as an edit of the photo (smoother and more even skin, a contour shadow under the jaw, darker and fuller brows, a defined lash line, …), ran the edited photo through the complete app pipeline (MediaPipe landmarks and blendshapes, all features, the CNN, fusion), and compared the PSL with the unedited photo (Table 3).

Table 4: SCUT-FBP5500 official 60/40 split (test $n = 2200$). Target: mean rating of 60 raters (1–5).

Model	Pearson $r \uparrow$	MAE ↓	RMSE ↓
AlexNet [16]	0.8298	–	–
ResNet-18 [16]	0.8513	–	–
ResNeXt-50 [16]	0.8777	–	–
Ours: 15 classic ratios only (constrained)	0.675	–	–
Ours: all 143 features (unconstrained)	0.820	–	–
Ours: all 143 features (constrained)	0.804	0.314	0.404
Ours: MobileNetV3 CNN	0.914	0.210	0.278
Ours: fused (deployed)	**0.920**	**0.204**	**0.268**

Fuller, darker brows, smoother skin and a defined lash line raise the score for most faces. Several popular suggestions do not: cheekbone contour, lip colour, under-eye brightening and stubble *lowered* the average score. The app therefore offers an action only if it raised the score for at least 60% of test faces of that sex, and lists the rest as "tested but didn't raise the score". For one person the predicted gain is $(w_{\text{cnn}}\,\Delta_{\text{cnn}} + w_{\text{geo}}\,\Delta_{\text{geo}})/\sigma$, where Δ_{cnn} is the measured CNN effect adjusted linearly for the person's own value of the action's main measurement, and Δ_{geo} is the feature model's response when the measured average feature change is applied to the person's own values. Each suggestion also lists the person's measurements that the action moves toward the top-rated average, and a tip specific to the worst of them. Fat loss, haircuts and expression cannot be simulated on a photo, so they appear separately, without a prediction, and only when the person's measurements point to them. Structural weaknesses (for example nose width) are named as not covered, because the plan never suggests surgery, injectables or UV tanning.

4.9 Interface and privacy

The app is built with React and TypeScript on Vite. The toolbar has only two inputs: a camera picker for the live feed and an Upload photo button. The video is drawn on one canvas with a deliberately light overlay: thin corner marks, a scan-progress line, and one small label with the PSL and tier. The side card shows the score, four pillar bars and three strengths and three things to improve; the full measurement list and the model weights are one click away rather than on screen by default. All model files and wasm runtimes are served from the app itself, so it also works offline on demo day. No frame or uploaded photo is ever sent over the network. In study builds (`?study`), Record saves a log file locally. Record logs *numbers only* (per-person PSL, model outputs, yaw, fps). An opening screen asks for confirmation that everyone in frame is 18+ and has consented.

4.10 Replicating the system

All code is in our repository. `README.md` lists the nine commands that rebuild every model and number in this paper from the public dataset, and `npm run dev` starts the app. Every number in this document is generated from the result files by `make_figures.py`.

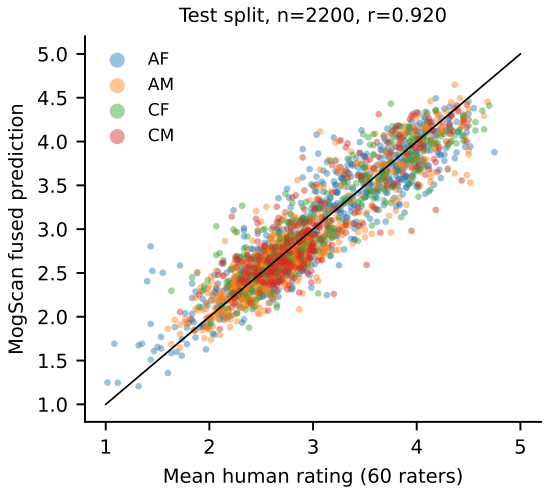


Figure 3: Fused prediction vs. mean human rating on the test split, colored by subgroup (A/C = Asian/Caucasian, F/M).

5 Evaluation

5.1 Agreement with human raters

Table 4 and Figure 3 show the results. The fused model reaches $r = 0.920$ on the held-out test split. That is higher than all three of the dataset paper's 60/40 baselines, although it is below recent large research models (≈0.92 [2]), which are not designed for real-time use in a browser. Expanding the explainable model from the 15 classic ratios ($r = 0.675$) to all 143 inputs raises it to $r = 0.804$, close to the unconstrained fit ($r = 0.820$), so the sign constraints that keep the explanations honest cost little. It is still weaker than the CNN, which sees everything at once, but it now carries enough independent signal to earn a larger fusion weight (0.13). The sex head reaches 99.1% accuracy. On the PSL scale the mean absolute error is 0.29 points, the exact tier matches in 62.4% of test faces, and the tier is within one step in 98.1%.

5.2 What each data source adds

Table 5 and Figure 4 refit the explainable model on each feature group alone and on everything *except* each group. Whole-face shape is the strongest single group, and proportions, jaw and expression also carry a lot of signal on their own. Most groups overlap heavily, so removing any one costs little; colouring & contrast and skin are the groups whose information no other group duplicates.

5.3 Do looksmax rules hold up?

Figure 5 shows the 25 features with the strongest raw correlation with the human rating. The strongest single feature is *Jaw width : face length* ($r = -0.51$, averaged across sexes). Several community rules do *not* hold up for these raters:

- **Canthal tilt**, the best-known looksmax metric, has essentially no correlation ($r \approx 0.00$). Before frontalization it seemed to matter ($r \approx 0.28$), but that came from head pose.
- **"Hunter eyes":** more *open* eyes go with higher ratings ($r \approx 0.37$).

Table 5: Ablation of the explainable model (test Pearson r). "Without" is the change in r when the group is removed.

Group	Features	Alone	Without
Whole-face shape	14	0.705	-0.001
Proportions	8	0.643	-0.000
Jaw & chin	8	0.639	-0.000
Expression	51	0.601	-0.002
Eyes	9	0.593	-0.006
Profile depth (3D)	10	0.573	-0.000
Colouring & contrast	8	0.547	-0.019
Mouth & lips	7	0.539	-0.000
Brows	5	0.534	+0.000
Nose	5	0.380	+0.001
Skin	6	0.378	-0.007
Leanness	3	0.308	+0.000
Hair & grooming	3	0.298	-0.002
Symmetry	6	0.274	+0.000

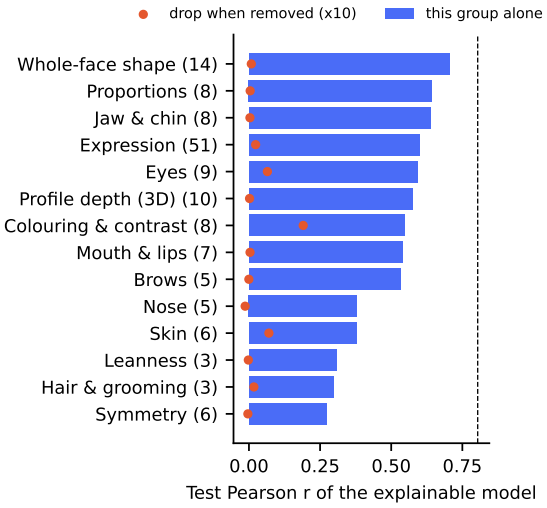


Figure 4: Test r of the explainable model using each feature group alone (bars) and the drop in r when that group is removed from the full model (×10, dots). Dashed line: all groups.

- **Wide jaw and high FWHR:** faces that are *narrower* in the jaw relative to the cheekbones ($r \approx -0.38$), with lower FWHR ($r \approx -0.26$) and softer projected jaw angles ($r \approx 0.36$), are rated higher. That is a V-shaped face, the opposite of the "wide, square jaw" ideal.
- **Symmetry** looks *negatively* related to attractiveness on the full dataset, but the effect disappears ($r \approx 0.00$) for faces within 3° of frontal. It comes from residual pose, since head-turned photos received higher ratings ($r = 0.28$ for |yaw|, male faces).

A longer chin relative to the philtrum ($r \approx 0.19$) and wider-set eyes ($r \approx 0.19$) agree with community advice. These results describe one dataset and one rater panel, and East Asian beauty norms may favour V-shaped faces. Still, they show why MogScan displays the *data-derived* target next to the community range instead of repeating forum rules, and why pose control is necessary for any measurement-based rating.

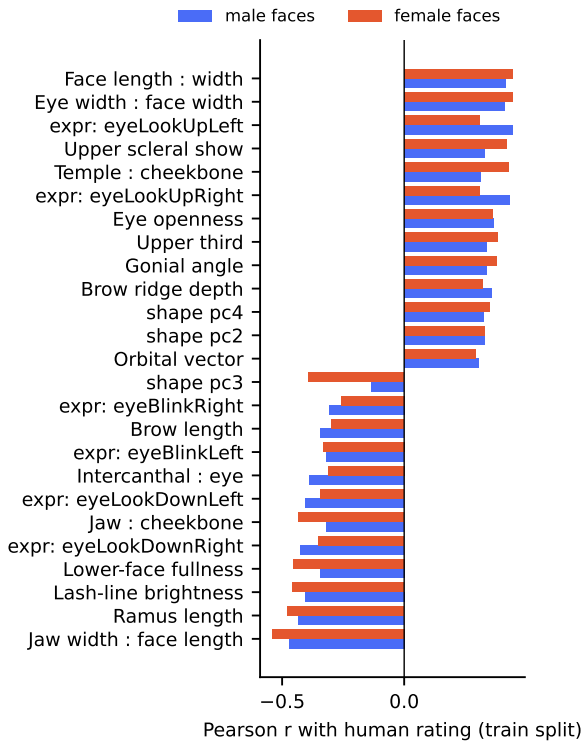


Figure 5: The 25 features most correlated with human ratings (training split), split by sex.

Table 6: Fused model by subgroup (test split). Signed bias = mean(pred − human).

Group	*n*	*r*	MAE	Signed bias
Asian female	792	0.922	0.212	-0.009
Asian male	811	0.930	0.179	-0.028
Caucasian female	285	0.906	0.232	+0.006
Caucasian male	312	0.886	0.220	-0.012

5.4 Subgroup bias

Table 6 breaks the error down by the dataset's four subgroups. SCUT-FBP5500 has only Asian and Caucasian faces, rated by one panel of 60 young raters. **People of other ethnicities are not represented at all**, so the model's behaviour on them is unvalidated. That is a serious limitation for a consumer product. The app's opening screen says so, and collecting a more diverse dataset is our first item of future work.

5.5 User study

[The user study has not been run yet. Follow study/protocol.md with 8–15 participants, export the Google Form to study/responses.csv and the session logs to study/logs/, run analyze_survey.py, analyze_logs.py and make_figures.py, then rebuild this PDF. Every number in this section is filled in automatically. Do not type numbers by hand.]

5.6 Performance and stability

[Fill from study logs (median PSL std. while still, median fps).] In our own development testing (Chrome on an Apple-silicon MacBook, photo mode, one face) the full pipeline ran at about 27–36 frames per second.

6 Ethics, Risks and Limitations

Rating faces can hurt people. Parental-safety reviewers rate PSL apps 18+ and warn that they can reinforce body dissatisfaction [1]. Our mitigations:

- an 18+ and consent screen;
- on-device processing with no storage;
- no competitive features: no rankings, duels or leaderboards between people;
- honest framing in the app ("a model's estimate of how one panel judged faces, not your worth");
- a wellbeing item in our survey.

Technical limitations:

- SCUT faces are studio-like frontal photos, while webcam frames vary in lighting and lens distortion, so there is domain shift;
- the mesh starts at the upper forehead, so true facial thirds and full face length are approximations;
- the jaw angle is a 2D projection, and depth features come from a monocular mesh, so they approximate a true profile;
- pixel features depend on lighting and camera; they were learned on studio photos;
- the dataset covers two ethnic groups and only binary sex labels.

Commercial use would require licensed training data.

7 Future Work

- **Diverse, licensed data:** collect consented ratings across many ethnicities, ages and rater groups, retrain, and publish subgroup error before any commercial launch.
- **Side-profile scan:** gonial angle, chin projection and the Ricketts E-line need a profile view. Prompt users to turn their head and fuse the measurements.
- **Progress tracking:** longitudinal, pose-normalized rescans that check whether the glow-up plan's predicted gains actually happen, so the plan can be recalibrated on real before/after data.
- **Body and presentation:** a pose model and full-body ratings would let us add the physique, shoulder-width and clothing factors that some looksmax checklists include.
- **Better CNN explanations:** Grad-CAM heatmaps for the texture cues the feature model cannot see, and a WebGPU backend for higher frame rates with six people in frame.

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